

Multicalibration for Unbiased Prevalence Estimation from Political Text

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Abstract

Political scientists increasingly use classifiers, and now large language models (LLMs), to estimate how prevalent concepts are in natural language, such as topics in legislative agendas or categories in open-ended survey responses. Standard validation with discriminative metrics (such as accuracy or AUC on a labeled sample) does not certify valid prevalence estimates nor transportability of the model to new contexts. Under covariate shift between the validation sample and the target sample, an accurate classifier can still produce badly biased prevalence estimates, and must be corrected for every new target. We show that *multicalibration*, calibration conditional on input features rather than on average, is sufficient for unbiased prevalence estimation under any covariate shift along the calibrated features. We demonstrate this property in a simulation as well as an exemplary application coding topics in the comparative manifestos corpus. We find that multicalibration produces estimates with near-zero bias under shift, where standard methods miss by a large margin.

Introduction

Machine learning classifiers, and increasingly large language models (LLMs), have become standard tools to estimate the prevalence of a concept in political text corpora. Researchers use them, for example, to code policy agendas across countries (Baumgartner et al. 2006; Benoit et al. 2026), detect protest and conflict events (Overos et al. 2024; King and Lowe 2003), score democratic institutions (Weidmann et al. 2026), categorize open-ended survey responses (Mellon et al. 2024), or label social media at scale (Gilardi et al. 2023; Ornstein et al. 2025). The quantity of interest is usually not the label on any single document but an *aggregate* count or prevalence (González et al. 2017; Hopkins and King 2010).

The standard way to validate such a classifier is to measure how well it discriminates: how closely its labels agree with human codes, reported as accuracy, F_1 , or AUC and judged against the intercoder reliability of the human benchmark (Halterman and Keith 2025; Hayes and Krippendorff 2007). For prevalence estimation this is not sufficient, for two related reasons. First, discrimination does not

imply an unbiased prevalence estimate even on the population the classifier was validated on. A classifier can rank documents almost perfectly yet misstate the proportion that are positive, because its false positives and false negatives need not cancel. The quantification literature corrects this, adjusting the raw count with error rates estimated on the labeled sample (González et al. 2017).

Second, that correction is tied to the population it was estimated on. When the same instrument is applied to a new target, for example to compare sub-corpora or track a quantity over time, it no longer holds in general. This is due to covariate shift. Across targets, the mix of documents changes (languages, venues, eras, authors) while the rule linking content to category stays fixed (e.g. through the codebook). When a target holds more of the cases the classifier misjudges, errors that cancelled on the validation sample no longer do. Classical quantifiers assume the opposite. Under label shift, documents of a category look the same everywhere and only their overall frequency changes, which rarely holds across languages and venues. A high AUC does not protect against this, and standard validation will not reveal it (Adcock and Collier 2001).

Existing tools each address part of the problem, but none yields a *reusable* instrument. Quantification methods estimate prevalence from imperfect classifiers (González et al. 2017). Classify & Count, its bias-corrected relatives (Adjusted Count / Rogan-Gladen (Rogan and Gladen 1978), the Saerens-Latinne-Decaestecker EM algorithm (Saerens et al. 2002), distribution-matching (Maletzke et al. 2018)), and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023) all assume label shift, and so do not transport under covariate shift. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) debias an estimate using gold labels drawn from the target itself. Both are target-specific, because each new population needs its own sample. Importance weighting (Shimodaira 2000; Sugiyama et al. 2007) applies under covariate shift, but its density ratio must be re-estimated for each target. A related literature on learned proxies (Knox et al. 2022) and on the reliability of LLM annotations (Bisbee et al. 2024; Argyle et al. 2023; Rodriguez and Spirling 2022) raises the same worry that imperfect measures distort inference. Weidmann et al. (2026) in particular find that LLM classifiers deviate from experts in systematic, feature-dependent directions, which conditioning on features can correct.

What ties a classifier to prevalence is calibration. The simplest estimator just averages the classifier’s scores, and that average is unbiased precisely when the classifier is calibrated, that is, when its predicted probabilities match empirical frequencies in the population at hand. Calibration, however, is a property of one population and need not survive a shift. Empirically it degrades under language and task shift even when accuracy is preserved (Yang et al. 2023; Ren et al. 2025). Confidence elicitation (Tian et al. 2023; Kadavath et al. 2022; Wang et al. 2023) and post-hoc recalibration (Wu and Resnick 2024) aim only at this global calibration, and are fragile in the same way. We show that *multicalibration*, calibration conditional on the input features rather than on average (Hébert-Johnson et al. 2018; Detommaso et al. 2024), fills this gap. A multicalibrated classifier is stable under reweighting of those features. Calibrated once on a labeled source sample, it returns unbiased prevalence estimates on any target whose shift the features capture and that lies within their support (Kim et al. 2022), with no target-specific re-estimation. We develop this into a measurement approach for political text and show which correction suits which shift regime. In a simulation and an application coding topics across countries and languages, multicalibration holds near-zero bias under shift, where the standard defaults miss by several percentage points.

Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a predictor $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, estimated from unlabeled target data $\{X_i\}$ and h . We assume *covariate shift*, which means that $P(X)$ differs between calibration and target while $P(Y | X)$ is stable, i.e. the coding rule stays fixed. Covariate shift is the complement of the *label-shift* assumption, which instead holds $P(X | Y)$ fixed while the prior $P(Y)$ moves.¹ Under concept drift, where $P(Y | X)$ itself changes across contexts, no purely statistical method recovers the target prevalence without labels from the target.

A predictor is *globally calibrated* if $\mathbb{E}[Y | h(X) = p] = p$ for all p . Then $\mathbb{E}[h(X)] = \pi$, and averaging its predictions is unbiased. But calibration on one population need not hold on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the predictor’s mean signed error in G . Global calibration requires only $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the prevalence bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

nonzero in general unless every $\epsilon_G = 0$.² A predictor can be perfectly calibrated on the validation sample yet badly biased on a target that reweights those groups. High classification accuracy or discriminative performance does not guard against this problem. Discriminative performance depends only on the ranking of scores, and is therefore invariant to the group errors ϵ_G that produce the bias.

Existing corrections fail because none removes the group errors ϵ_G themselves. Output-based corrections (Classify & Count, Rogan-Gladen, global isotonic recalibration (Wu and Resnick 2024)) estimate aggregate positive and error rates on the source; these are weighted averages over groups, and they no longer cancel once the target reweights the groups. Label-shift quantifiers (SLD/EMQ (Saerens et al. 2002), distribution-matching, the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023)) fail for a different reason. They carry no per-document ϵ_G , but the identity they solve presumes $P(X | Y)$ is stable.

Multicalibration drives *every* ϵ_G to zero, not merely their weighted sum, by replacing h with a predictor f calibrated within each group. Formally, f is multicalibrated with respect to a class of groups $\mathcal{G}$ if $\mathbb{E}[Y | f(X) = v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018).³ A multicalibrated predictor then satisfies the Kim et al. (2022) “universal adaptability” guarantee. For any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$,

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*,$$

because the density ratio is constant within each $\mathcal{G}$ -cell, so the within-cell $\epsilon_G = 0$ carries through. One instrument, calibrated once, is unbiased on every target whose shift lies along the calibrated features and within support, with no knowledge of the shift and no target-specific estimation.

¹The heuristic that stable $P(Y | X)$ is natural when features cause the outcome, and stable $P(X | Y)$ when the outcome causes the features, traces to @scholkopf2012causal; we rely on the invariance rather than on adjudicating the causal direction.

²Because ϵ_G is estimated on the source, the identity is exact when each group is homogeneous enough that the density ratio is constant within it, and a first-order approximation otherwise; see SI Section S1.

³A slightly weaker property, *multi-accuracy* (zero average error within each group), already suffices for unbiased prevalence; multicalibration is stronger but the better practical target (SI Section S1.8).

Importance weighting uses the same density ratio, reweighting the labeled source to match the target (Shimodaira 2000; Sugiyama et al. 2007). It is valid under the same assumption, but it corrects the *sample* rather than the *predictor*. The density ratio must therefore be re-estimated for each target, and the method breaks under positivity violations. Multicalibration instead folds the correction into the predictor once, and so matches per-target importance weighting without per-target work.

The guarantee rests on two conditions. First, the target’s feature values must lie within the calibration support. This condition can be tested from the unlabeled target data. Second, the multicalibration features must be expressive enough to capture every dimension along which the shift interacts with the outcome. Expressiveness is an ignorability condition, which is not verifiable from data. It rests on the same substantive judgment as any observational adjustment. So does the covariate-shift assumption itself. Whether a shift is covariate rather than label shift or concept drift cannot be read off unlabeled data and must be argued from how the concept is defined. These conditions do not make multicalibration perform worse than the usual corrections. To the contrary, since a multicalibrated predictor is also globally calibrated, and enlarging $\mathcal{G}$ tends to reduce the residual bias. In practice we include every metadata dimension plausibly related to both the outcome and selection into a target, and where coarse cells might hide content that drives Y , text-derived features such as embeddings can be added (Rodriguez and Spiraling 2022). We obtain the multicalibrated predictor with MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

The bias reduction rests on two conditions. First, the target’s feature values must lie within the calibration *support*. This is testable from unlabeled target data. Second, the feature class $\mathcal{G}$ must be *expressive* enough to capture every dimension along which the shift interacts with the outcome. Expressiveness is an ignorability condition, which is not verifiable from data. It rests on the same substantive judgment as any observational adjustment. So does the covariate-shift assumption itself. Whether a shift is covariate rather than label shift or concept drift cannot be read off unlabeled data and must be argued from how the concept is defined. However, since a multicalibrated predictor is also globally calibrated, and enlarging $\mathcal{G}$ tends to reduce the residual bias. In practice we include every metadata dimension plausibly related to both the outcome and selection into a target, and where coarse cells might hide content that drives Y , text-derived features such as embeddings can be added (Rodriguez and Spiraling 2022). We obtain the multicalibrated predictor with MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

Simulation

We first illustrate the mechanism in a setting simple enough to verify by hand. We generate synthetic data with a binary covariate $X \in \{0, 1\}$ and outcome Y with $P(Y = 1 \mid X = 1) = 0.85$ and $P(Y = 1 \mid X = 0) = 0.15$. A classifier produces deterministic, systematically biased scores (a 10% multiplicative bias within each stratum, so $\hat{p} = 0.135$ when $X = 0$ and $\hat{p} = 0.935$ when $X = 1$), wrong within each stratum but corrected to the true prevalence by a single global recalibration step on the balanced training distribution $P(X = 0) = 0.5$. We then estimate prevalence on target distributions with $P(X = 0)$ ranging from 0.01 to 0.99, holding all calibration parameters fixed at their training values, over 50 replications.

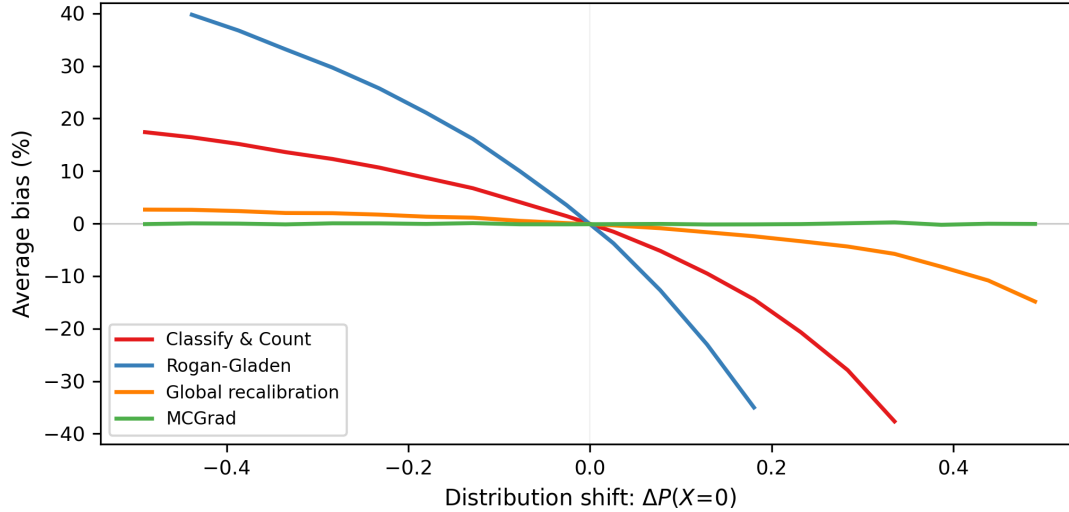


Figure 1. Relative prevalence bias under covariate shift (50 runs). The x-axis is the change in $P(X = 0)$ from the training value of 0.5. Classify & Count and Rogan-Gladen diverge with shift; global recalibration shows moderate bias; the multicalibrated estimator stays near zero. Cropped at $\pm 40\%$; the full range, additional methods, and RMSE are in SI Sections S4–S5.

At the training distribution ($\Delta P(X = 0) = 0$) every method is approximately unbiased (Figure 1). As the target shifts, they diverge. Classify & Count and Rogan-Gladen grow with the shift, the latter’s ratio form amplifying the error beyond $\pm 40\%$. Global recalibration shows moderate but nonzero bias, up to roughly 15% at the most extreme shift. Only the multicalibrated estimator stays near zero throughout. Every estimator in Figure 1 differs only in how it turns the classifier’s scores into a prevalence; all share the same ranking of X , and so have identical discriminative performance.

Example: LLM coding of political text under covariate shift

Next we demonstrate the mechanism on a more realistic application.⁴ We use Claude Opus 4.6⁵ to classify 30,000 Comparative Agendas Project texts (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12). The documents span six sub-populations across four countries and languages, 5,000 each: Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles, and Belgian TV news. Law & Crime is a rare topic, coded in 7.9% of documents at baseline and 6–20% across the shift scenarios. The design reproduces the researcher’s situation under controlled conditions. Because the published expert codes give the true prevalence in every sub-population, we construct targets with a known type and size of covariate shift and measure the resulting bias exactly.

We compare two output modes spanning current practice: the discrete Yes/No classification used in essentially all applied studies, and a probability score from direct elicitation of $P(\text{Yes})$, the confidence-elicitation recommendation (Tian et al. 2023; Kadavath et al. 2022). Discrimination is strong in both (AUC 0.960 binary, 0.987 scores; per-language 0.983–0.994). We draw the calibration

⁴An American Community Survey employment benchmark with exact ground truth (SI Section S3) shows the correction is not specific to text or to language models but holds for an ordinary classifier.

⁵We replicate the results with an open-weights model Llama 3.3 70B (SI Section S2) which confirms the results reported and allows full replication of our analysis.

set ($n \approx 13,400$) equally from four sub-populations, and hold out two others (Spanish media, Belgian TV) as out-of-support targets. These share country and language with the calibration data but add an unseen document type. We calibrate MCGrad on observable metadata only: country, document type, party, decade, and document length. These features serve as the group class $\mathcal{G}$. For the binary condition, MCGrad receives the Yes/No label as a categorical feature, with scores initialized at the calibration base rate, and never sees a probability. For the score condition, it receives $P(\text{Yes})$. We benchmark against Classify & Count, Rogan-Gladen, isotonic regression, importance weighting (IPW), and SLD (SI Table S2).

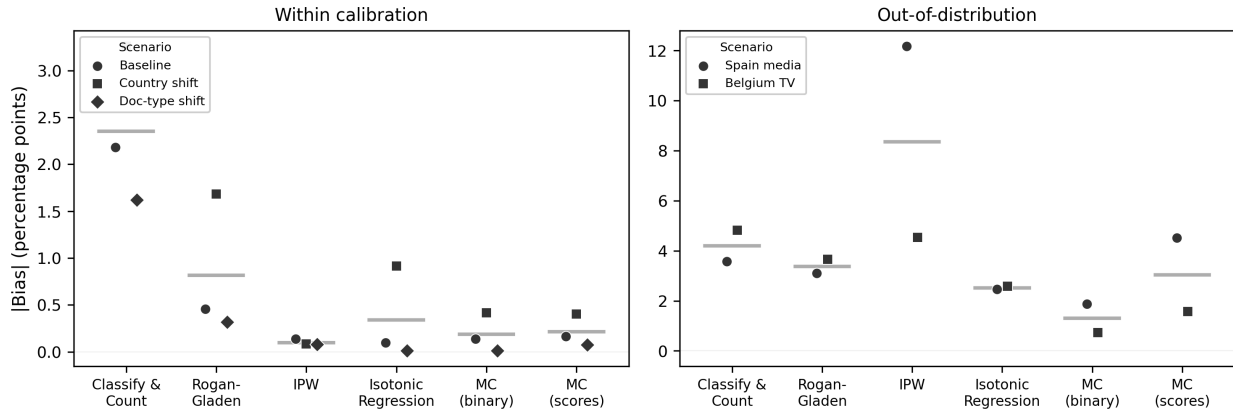


Figure 2. Absolute prevalence bias (percentage points) across the shift gradient: a no-shift baseline, two within-support shifts (country and document-type composition), and two out-of-support targets. Marker shape denotes scenario; horizontal lines are scenario means. The two multicalibration variants (MC binary on Yes/No labels, MC scores on elicited probabilities) are indistinguishable within support (both $\leq 0.4\text{pp}$, below the 0.4–0.6pp sampling error in the gold base rate). Full numbers in SI Table S2.

The classifier clears any discriminative-validation bar, with per-language AUC of 0.98–0.99, yet Classify & Count, the applied default, is biased by +1.6 to +4.8pp across scenarios (Figure 2), a relative overstatement of nearly 30% at baseline and over 40% at its worst. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. Isotonic regression is good within support ($\leq 0.9\text{pp}$) but worse out of it (2.5–2.6pp). SLD, the label-shift quantifier, is already biased at baseline (+7.4pp, since the elicited scores are not the calibrated posteriors its EM step assumes) and reaches +21.6pp out of support. This confirms that the label-shift family does not transport under covariate shift.⁶

Multicalibration on the bare Yes/No labels achieves near-zero bias within calibration ($\leq 0.4\text{pp}$) and degrades gracefully out of support (+0.7pp on Belgian TV, -1.9pp on Spanish media), from the label and metadata alone. On probability scores it matches this within support ($\leq 0.4\text{pp}$) and is somewhat worse out of support (-4.5pp on Spanish media, +1.6pp on Belgian TV). Bare labels match or beat elicited probabilities, because the correction comes from the metadata features and the calibration sample, not from the quality of the model’s confidences. IPW is near-zero within support but fails out of support (-12.2pp on Spanish media), where the target contains feature values absent from the source (a positivity violation). It must also re-estimate density ratios for

⁶ReadMe [hopkinsking2010nonparametric], the canonical political-science quantifier, is classifier-free (it estimates the proportion from raw text rather than correcting the LLM’s outputs), and its bias runs off this scale; we report it in SI Section S7 (Table S4). It fails as the label-shift diagnosis predicts, near-zero only without shift and exceeding +60pp on some shifts; ReadMe2 [jerkzakingstrezhnev2023improved] shares the assumption and cannot escape it.

each target.

Discussion

Prevalence estimation across populations is a common goal in political science research. Discriminative validation shows only that a classifier separates classes. It is silent on whether the resulting prevalence estimates are unbiased, both in the population where discrimination was validated and in any other. In our application a classifier with per-language AUC above 0.98 still produced default prevalence estimates off by roughly 30–40% of the true rate, already without shift and worse under it, and would have passed any discriminative check while misreporting prevalence. Multicalibration, calibration conditional on features rather than on average, removes this bias where discrimination was validated and keeps it removed as the target shifts.

However, the guarantee has limits. It holds only along calibrated dimensions, so calibration data must span the anticipated variation. It also requires labeled calibration data, which in zero-shot settings reintroduces some hand-coding, though far less than per-document labeling and only once. Under mixtures the covariate-shift condition holds only approximately. Our applications have ground truth, so we can verify the estimates directly; a practitioner usually cannot, since the target’s labels are what is missing.

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